Iteration logic (loops) in C#

When you want to repeat a block of code more than once, then you can use loop constructs. This article explains the For loop, While loop, Do..While loop and the Foreach loop.

FOR loop

Use a For loop when you know exactly how many times you want to repeat a loop. It will iterate a fixed number of times.

The For loop contains a 3-part expression inside round brackets:

- The first part declares an integer variable and assigns it an initial value. In the example below, the variable is called counter, but you can call it anything you like.
- The second part is the condition that is checked each time the loop repeats. If that condition evaluates to true then the loop is repeated another time, otherwise it stops.
- The third part adds one to the counter so that it changes on each iteration (so that the loop will eventually stop). The ++ symbol means add one to the value of the variable.

```
for (int counter=0; counter < 10; counter++) {
    Console.WriteLine("Iteration " + counter);
}
```

WHILE loop

The While loop repeats until the condition becomes false. The number of times it will repeat is not pre-determined. Inside the loop you must remember to change the variable that forms part of the condition (so that the loop will eventually stop). Here is the same example written with a WHILE loop:

```
int counter = 0;
   while (counter < 10) {
       Console.WriteLine("Iteration " + counter);
       counter++;
   }
```

DO…WHILE loop

This loop is the same as a while loop except that the condition is evaluated at the end of every iteration instead of at the start of every iteration. A Do..While loop will always execute at least once.

```
int counter = 0;
        do {
           Console.WriteLine("Iteration " + counter);
           counter++;
        } while (counter < 10);
```

FOREACH loop

The Foreach loop is used for iterating through a collection. We will see examples later in this course.